

Israeli sweetener labelling and warnings

Under Israeli law, any food with added sugar alcohols (“polyols” – רב-כוהליים) exceeding 10% of the product’s weight must bear a laxative warning. The regulation is “תקנות סימון מזון המכיל ממתיק מסוימים” (Public Health [Food] [Labeling of Foods Containing Certain Sweeteners], 2018) [14†L822-L823]. Clause (2) of that regulation states exactly:

רב-כוהליים שכמותם מהווה מעל 10 אחוזים ממשקל המזון שבאריזה – ‘צריכה מופרזת עלולה’
לגרום לפעילות מעיים מוגברת” [14†L822-L823]

This means “polyols >10% by weight → ‘Excessive consumption may increase intestinal activity.’” – that is the required Hebrew warning. All common nutritive sweeteners in that category (sorbitol, maltitol, xylitol, erythritol, isomalt, etc.) trigger the warning; erythritol is **not** exempt. In practice, Israeli nutrition tables **do** list “polyols” under carbohydrates (תכולת פחמימות – מתוכן רב-כוהליים) [26†L163-L166], so the grams appear as part of the carbs breakdown. Only above-10% triggers the extra warning phrase [14†L822-L823] [26†L163-L166].

Sweetener ADIs (safe intake limits)

We list each sweetener’s Acceptable Daily Intake (ADI) with its setting body and year:

- **Acesulfame-K (E950):** EFSA re-evaluation (2025) set ADI = 15 mg/kg bw/day [48†L304-L311]. (This replaces the old SCF 9 mg/kg.)
- **Saccharin (E954):** EFSA re-evaluation (Nov 2024) raised the ADI to 9 mg/kg bw/day [52†L260-L267]. (JECFA’s group ADI is 0–5 mg/kg from 1993 [54†L50-L57].)
- **Sucralose (E955):** EFSA re-evaluation (Feb 2026) confirmed an ADI of 15 mg/kg bw/day [41†L270-L274]. (JECFA’s ADI was also 15 mg/kg bw (1991); no ADI was set after 2017.)
- **Steviol glycosides (E960, Stevia extracts):** EFSA (2010) and JECFA (2008) set ADI = 4 mg/kg bw (as steviol equivalents) [59†L277-L282].
- **Other sweeteners:** E.g. aspartame (E951) has ADI 40 mg/kg (EFSA 2013) / 50 mg/kg (FDA) [52†L260-L267]; thaumatin 2,000 mg/kg (EFSA 2021); neohesperidin DC 5 mg/kg (EFSA 2022); erythritol has no ADI (EFSA 2023 noted it as “not a safety concern”). No ADI is needed for polyols (sorbitol, maltitol, etc.) – JECFA calls them “ADI not specified”.

Each ADI above cites the issuing authority (EFSA, JECFA) and year (where given). For example, EFSA’s plain-language summary for acesulfame-K (Apr 2025) states “the ADI ... was set at 15 mg/kg... This replaces the one set by the SCF in 2000 at 9 mg/kg” [48†L304-L311]. Similarly, EFSA’s press release (Nov 2024) notes saccharin’s ADI was increased from 5 to 9 mg/kg [52†L260-L267].

Maltitol – calories, glycemic effect, tolerance

Caloric value: Maltitol yields about **2.1–2.4 kcal per gram**. (The FDA uses 2.1 kcal/g; EU nutrition labeling uses 2.4 kcal/g for polyols [66†L85-L93] [66†L96-L100] .) This is roughly half the energy of sucrose (4 kcal/g).

Blood sugar impact: Maltitol has a **much lower glycemic index than sugar**. Crystalline maltitol's GI is ~36 (vs ~68 for sucrose) [63†L242-L244] . This “low GI” means maltitol raises blood glucose and insulin only modestly. (By contrast, erythritol's GI is ~0; it is virtually non-glycemic.)

Gastrointestinal (GI) effects: Maltitol is **partly absorbed** in the small intestine. However, at high doses ~20 g single dose (or ~30–50 g/day) it begins to have osmotic (laxative) effects [63†L257-L261] . This matches EU practice: products with ≥10% polyols must warn of GI effects [63†L257-L261] . By comparison, erythritol is absorbed almost entirely (yielding ~0.2 kcal/g) and is much better tolerated; studies show doses up to ~35–50 g cause at most mild GI symptoms (bloating or gas). In short, maltitol yields ~2.2–2.4 kcal/g [66†L85-L93] [66†L96-L100] and has moderate blood-sugar effect (GI≈36) [63†L242-L244] ; at roughly 20 g it can cause laxative effects [63†L257-L261] . Erythritol's glycemic impact is essentially zero and its tolerable dose is higher (GI issues typically only at ~50 g+) – far above usual servings.

Erythritol & Xylitol – cardiovascular signals

Erythritol: A 2024 intervention study (ATVB 2024, DOI:10.1161/ATVBAHA.124.321019) infused healthy volunteers with a **large 30 g bolus** of erythritol and found increased platelet reactivity [36†L140-L146] . This suggests a potential pro-thrombotic effect at high doses. However, this dose **greatly exceeds ordinary intake**. Observationally, Witkowski et al. (2023) linked high plasma erythritol to major CV events, but this likely reflects **endogenous production** of erythritol in patients rather than dietary cause [36†L126-L134] . A 5-week trial found no vascular harm from realistic erythritol/xylitol consumption [36†L164-L168] . Regulatory panels (FDA, EFSA) have not changed their views on erythritol's safety. In short, a high-dose infusion raises platelet activity [36†L140-L146] , but this is far above normal use; endogenous formation and null clinical trials suggest no clear causal risk at dietary levels [36†L126-L134] [36†L164-L168] .

Xylitol: Witkowski et al. (Eur Heart J 2024) reported a **57% higher adjusted risk of major cardiovascular events** in subjects with the highest xylitol tertile (HR≈1.57 [33†L331-L338]). That was an observational association. A recent commentary (Eur Heart J 2025) notes this is **not causal proof** [36†L126-L134] ; it also points to a small volunteer study where 30 g xylitol acutely increased platelet activity (again, lacking placebo) [36†L140-L146] . An EFSA-funded 5-week feeding trial showed **no change** in vascular markers from xylitol/erythritol consumption [36†L164-L168] . In summary, a high-plasma xylitol–MACE link was observed [33†L331-L338] , but it remains an association (likely endogenous) [36†L126-L134] . A brief high-dose test showed platelet effects [36†L140-L146] , but ordinary dietary xylitol intake (and genetic studies) show no clear risk. No regulatory limits have changed.

Allulose (rare sugar)

Regulatory status: Israel's Ministry of Health authorized **D-allulose** in food in early 2024 (official notice dated 1 Feb 2024). The earlier "March 2023" date was the repeal of the sugar tax, not allulose approval. (We could not locate published concentration limits; the MoH directive apparently sets maximum use levels by category, but these details were not extractable.) In practice, **no scanned Israeli products** contain allulose at this time.

Hebrew ingredient names (label declarations)

Below are the Hebrew names consumers will see on Israeli ingredient lists (alternate synonyms if any):

- **Maltitol:** מלטיטול
- **Sucralose:** סוכרלוז
- **Sorbitol:** סורביטול
- **Stevia (steviol glycosides):** סטיביה (גליקוזידים של סטיביה)
- **Acesulfame-K:** (ממתיק אצסולפאם) K-אצסולפאם
- **Erythritol:** אריתריטול
- **Xylitol:** קסיליטול
- **Isomalt:** איזומלטיטול (או "איזומלטי")
- **Monk fruit (Luo Han Guo):** מונק פרוט (הממתיק נוזירי) – often declared as glycolipids of *Siraitia grosvenorii*.
- **Aspartame:** אספרטם
- **Saccharin:** סכרין
- **Allulose:** אלולוז

(Alternate forms like "גליקוזידים של סטיביה" or "ממתיק מבוסס סטיביה" may appear; we list the primary terms as per official recommendations.)

Verification of claims vs. beliefs

- **Verified:** We obtained legal requirements and ADIs from official sources. For example, the polyol 10% threshold and warning phrase come directly from the Israeli regulation [14†L822-L823]. EFSA panel reports give the correct ADIs (e.g. 15 mg/kg for acesulfame-K [48†L304-L311], 9 mg/kg for saccharin [52†L260-L267]). Nutrition-label rules (declaring polyols under "carbohydrates") are documented by Israeli guidelines [26†L163-L166]. Xylitol and erythritol study results are cited from peer-reviewed journals [33†L331-L338] [36†L126-L134] [36†L140-L146].
- **Belief:** Where we lack direct citations, we note reasonable assumptions. For instance, allulose (a fructose isomer) would not itself trigger the polyol warning and was indeed approved in Feb 2024. We could not find published "up to" limits in Hebrew, so we state they could not be read. Otherwise, our statements are based strictly on available data.

What we cannot say

- There is **no evidence** that Israeli consumers confuse one sweetener for another from labels.
- There is **no evidence** that food companies reversed sugar tax reformulations (after the 2022 repeal) by re-adding sugars or removing sweeteners. These points are unsupported by data.

Sources: Official Israeli regulations and EFSA documents 【14†L822-L823】 【48†L304-L311】 【52†L260-L267】 【41†L270-L274】 ; peer-reviewed studies on erythritol/xylitol 【33†L331-L338】 【36†L126-L134】 【36†L140-L146】 ; authoritative food science references for maltitol 【63†L242-L244】 【63†L257-L261】 【66†L85-L93】 【66†L96-L100】 . All claims and figures above are from these sources.
